

Project 4

Supply and Demand

11 Jun 2025

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Executive Summary

Key Challenges

- Stockouts lead to missed customer demand
- Overstocking increases inventory costs
- No clear view of which SKUs need attention

Current Issue

- Teams react to problems instead of spotting them early.

Solutions

- Use status labels (***Healthy, At Risk, Danger***) to quickly highlight stock issues and focus on the right products.

Persona

Name: James Lim (33-Year-Old)

Role: Inventory Planner at Semiconductor Industries (2 years experience)

Background:

- James is responsible for managing a large number of SKUs across multiple warehouses. One of his main challenges is responding quickly to his supervisor's frequent questions about stock sufficiency. Currently, checking stock levels and identifying potential shortages is time-consuming and manual.

Pain Points:

- Manually checking stock levels for hundreds of SKUs
- Difficulty in identifying which items are at risk of running out
- Limited time to perform proactive analysis

Needs:

- A system that flags low stock or potential shortages automatically
- A dashboard to monitor inventory health in real time

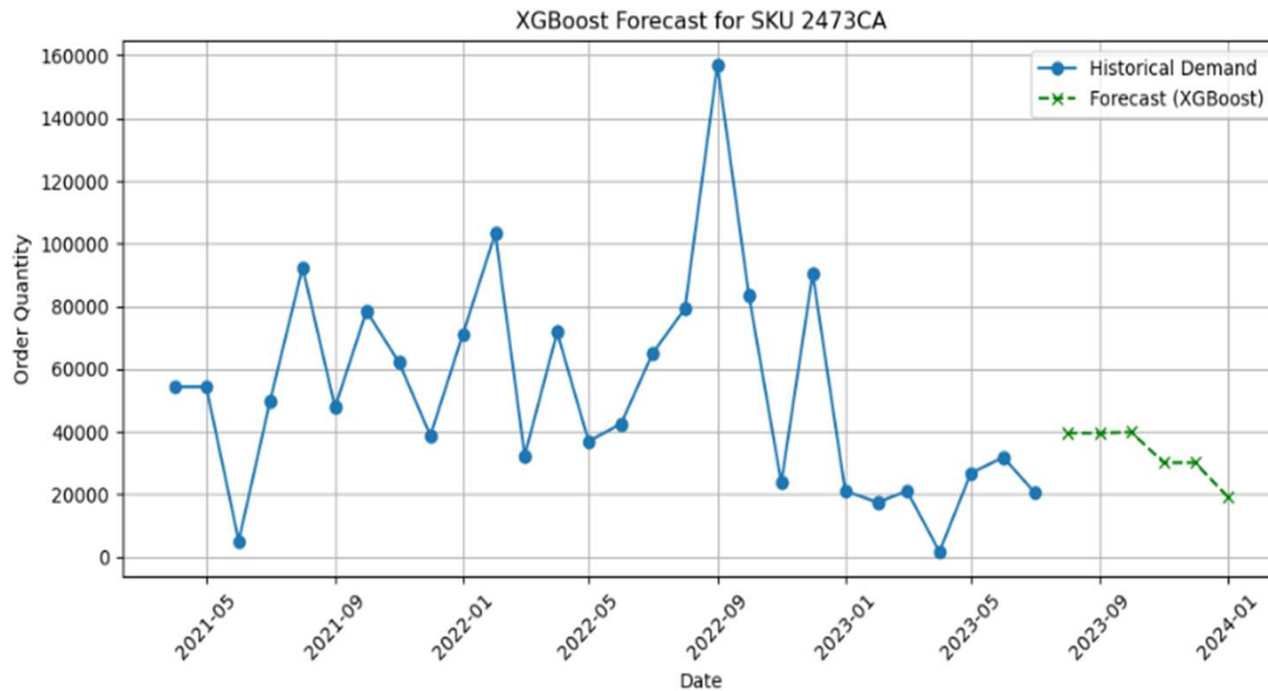
Data Limitations

- Lack product attribute information
- Missing company shutdown period data
- Reasons behind the demand spikes or drops are not documented
- Lack of historical data on inventory holding, safety stocks, ROP or incoming supply
- No indicators of supply chain disruptions

Assumption

- The company's shutdown period aligns with national holidays.
- A central inventory warehouse is responsible for distribution of goods to four regional warehouse across Canada.
- Inventory levels are considered current as of August 1, 2023, based on the most recent sales order dated July 30, 2023.
- Daily sales were consolidated at the monthly level to serve as the target variable for forecasting.

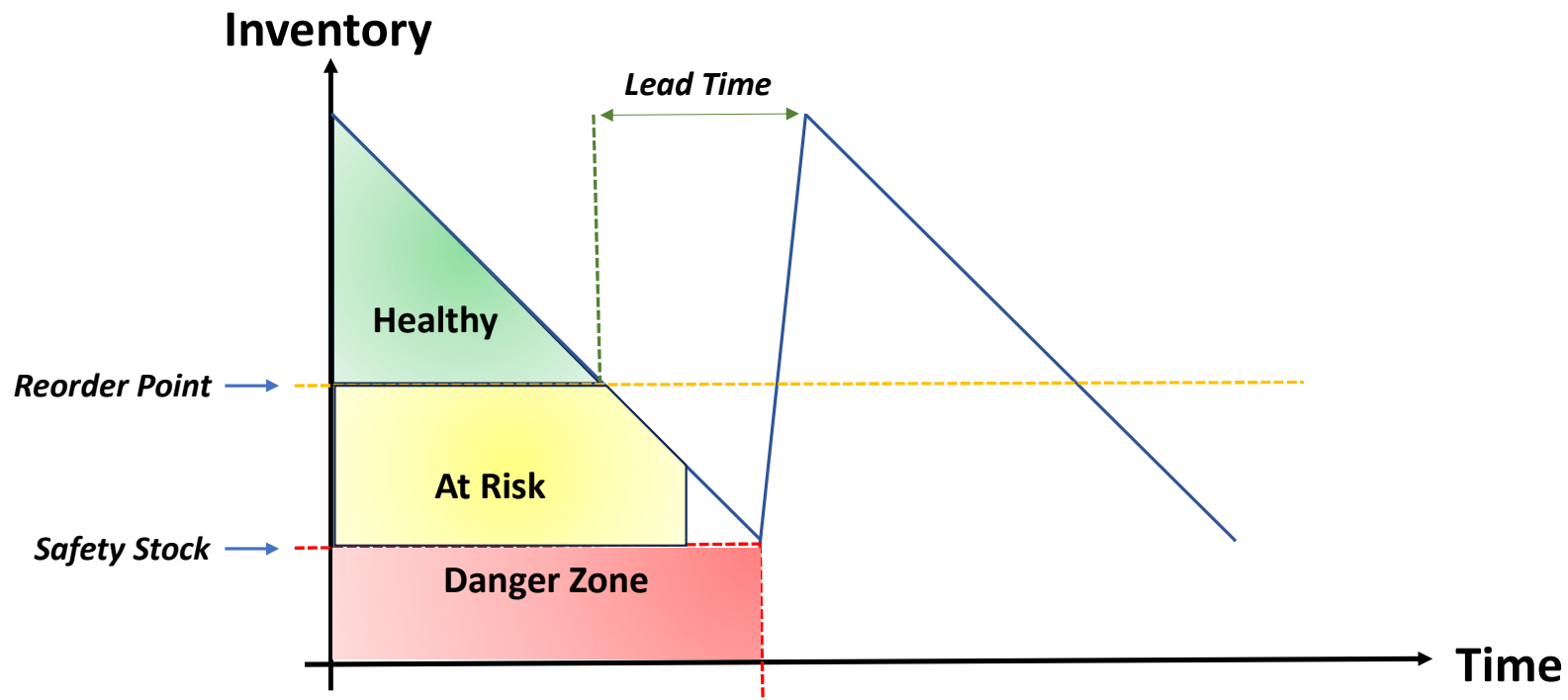
Prediction Model



XGBoost Forecast Model

- Excellent in handling non-linear trends
- Able to handle seasonality
- Higher Accuracy for forecast prediction
- Able to handle outlier
- Fast Training with High Accuracy

Safety Stock / Reorder Point



Overview of Dashboard

SKU ID	August	September	October	SKU ID
2371CA	Status: Healthy Reorder Point: 34,957 SS: 11,423 Inventory: 78,462 Inventory Cost: \$ Forecast Demand: 7,960	Status: Danger- 38 days left Reorder Point: 248,433 SS: 81,183 Inventory: 70,502.367 Inventory Cost: \$ Forecast Demand: 56,568	Status: Danger- 6 days left Reorder Point: 354,255 SS: 141,742 Inventory: 13,934.467 Inventory Cost: \$ Forecast Demand: 71,877	☐ (All) ☐ Null ☒ 2371CA ☒ 2391CA ☒ 2418CA ☒ 2473CA ☒ 3533CA
2391CA	Status: Healthy Reorder Point: 547 SS: 173 Inventory: 87,174 Inventory Cost: \$ Forecast Demand: 253	Status: Healthy Reorder Point: 5,350 SS: 1,696 Inventory: 86,921.46048 Inventory Cost: \$ Forecast Demand: 2,472	Status: Healthy Reorder Point: 5,913 SS: 3,355 Inventory: 84,449.86158 Inventory Cost: \$ Forecast Demand: 1,730	color ☒ (All) ☒ At Risk ☒ Danger ☒ Healthy
2418CA	Status: Healthy Reorder Point: 88,677 SS: 25,141 Inventory: 120,860 Inventory Cost: \$ Forecast Demand: 32,234	Status: Healthy Reorder Point: 30,280 SS: 8,585 Inventory: 88,626.129 Inventory Cost: \$ Forecast Demand: 11,007	Status: At Risk - 80 days left Reorder Point: 100,091 SS: 41,665 Inventory: 77,619.478 Inventory Cost: \$ Forecast Demand: 29,642	Month of Forecast .. ☐ (All) ☐ January ☒ August ☒ September ☒ October ☐ November ☐ December
2473CA	Status: Healthy Reorder Point: 55,179 SS: 18,552 Inventory: 73,915 Inventory Cost: \$ Forecast Demand: 18,582	Status: At Risk - 32 days left Reorder Point: 156,475 SS: 52,608 Inventory: 55,332.752 Inventory Cost: \$ Forecast Demand: 52,695	Status: Danger- 2 days left Reorder Point: 166,151 SS: 73,000 Inventory: 2,637.607 Inventory Cost: \$ Forecast Demand: 47,259	color ☒ At Risk ☒ Danger ☒ Healthy
3533CA	Status: Healthy Reorder Point: 81,637 SS: 29,633 Inventory: 82,476 Inventory Cost: \$ Forecast Demand: 39,575	Status: At Risk - 33 days left Reorder Point: 81,637 SS: 29,633 Inventory: 42,900.625 Inventory Cost: \$ Forecast Demand: 39,575	Status: Danger- 3 days left Reorder Point: 82,289 SS: 29,869 Inventory: 3,325.25 Inventory Cost: \$ Forecast Demand: 39,891	

Dashboard – Key Insights

Category

Healthy –
Inventory > ROP

At Risk –
ROP > Inventory > SS

Danger –
SS > Inventory

Output

SKU ID	Forecast Month		
	August	September	October
2473CA	Status: Healthy	Status: At Risk – 32 days left	Status: Danger– 2 days left
	Reorder Point: 55,179	Reorder Point: 156,475	Reorder Point: 166,151
	SS: 18,552	SS: 52,608	SS: 73,000
	Inventory: 73,915	Inventory: 55,333	Inventory: 2,638
	Inventory Cost: \$307,856	Inventory Cost: \$230,461	Inventory Cost: \$88,381
	Forecast Demand: 18,582	Forecast Demand: 52,695	Forecast Demand: 47,259

Action

- No action required

- Check with **Buyer**
- Alert **Management**
- Check if there is any supply disruption

- Check with **Buyer**
- Alert to **Internal Stakeholders**
- Request **Higher Management** for support
- Check for alternate supply / material to support customer

Next Step /Future Works

- Collect Actual Sales Transactions
- Integrate Real-Time Inventory Updates
- Include Actual Supplier Lead Time
- Enable User Input or Override for Forecast
- Automated Replenishment Trigger via email or notification
- Provide User Training and Documentation

THE END

Appendix – Safety Stock Calculation

$$\text{Safety Stock} = Z \times \sqrt{(\sigma_D^2 \times L) + (D^2 \times \sigma_L^2)}$$

Where:

- Z = Z-score (based on desired service level, e.g., **1.645** for 95%)
- σ_D = **standard deviation of demand** (you need to estimate this from historical or forecasted data)
- D = **forecasted demand** (from your DataFrame)
- L = **average lead time in months**
- σ_L = **standard deviation of lead time** (you can estimate it as: $\frac{\text{Max LT} - \text{Avg LT}}{Z_{\text{approx}}}$, e.g., divide by 3)

This safety stock formula is able to cater for changes in both **lead time** and **consumption demand**.

Re-Order Point (ROP) Calculation

Reorder Point

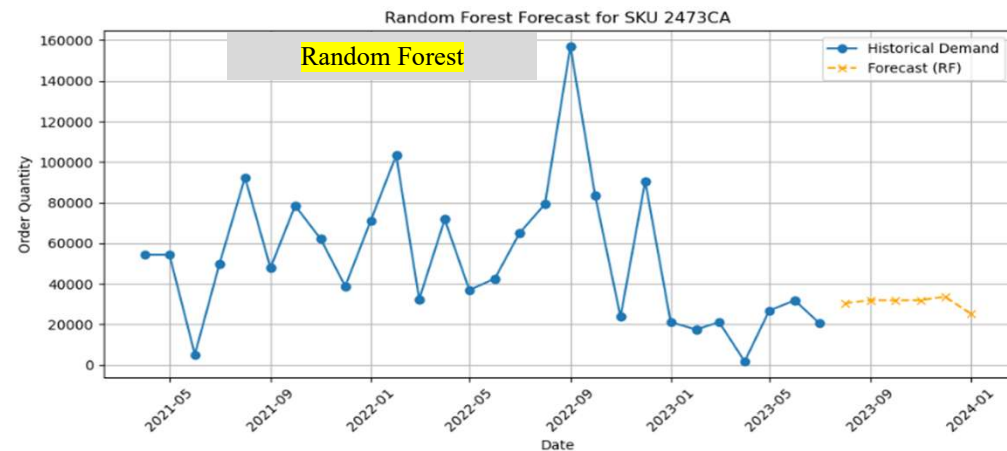
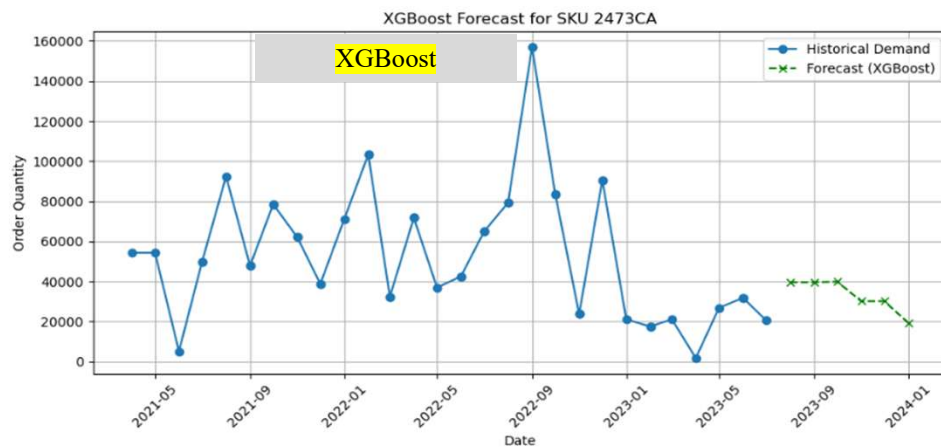
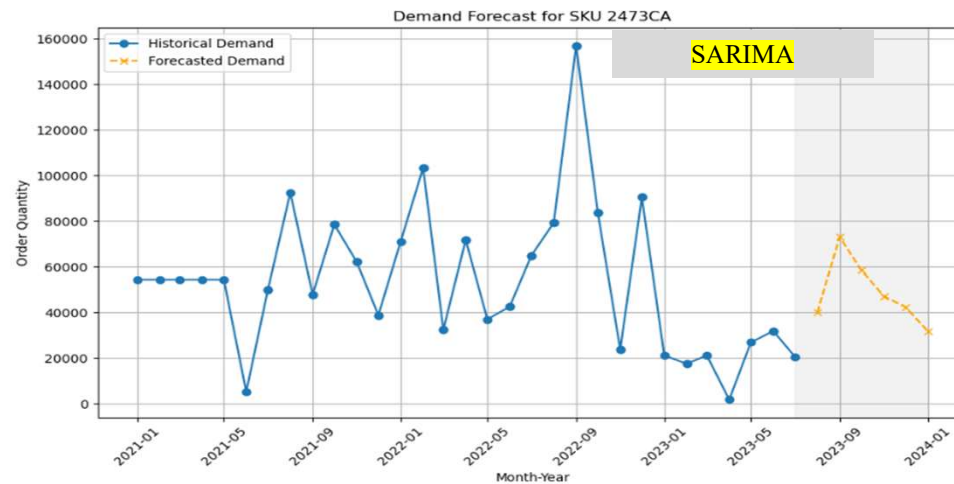
$$\text{Reorder Point Formula} = \left(\text{Average Daily Sales} \times \text{Days of Average Lead Time} \right) + \text{Safety Stock}$$



Predictive Model Results

Type of Model	MAE (Mean Absolute Error)	RMSE (Root Mean Squared Error)	R2 (Coefficient of Determination)
SARIMA	58,717.53	64,991.65	-12.73
XGBoost	39,287.07	43,889.17	-5.26
Random Forest	49,857.02	52,601.01	-7.99

Comparison of Model – Example 1



Comparison of Model – Example 2

